

Implementing High Availability with AWS Elastic Load Balancing and Target Groups

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1. Abstract

This document describes the process of deploying a highly available web application architecture on AWS, utilizing EC2 instances in multiple Availability Zones, an Application Load Balancer (ALB), and a Target Group. The solution ensures automatic failover and evenly distributes user traffic, maximizing resilience and minimizing downtime.

2. Introduction

High availability is a crucial requirement for modern web applications. By distributing resources across separate Availability Zones and implementing robust health checks and traffic management, this architecture minimizes the risk of outages caused by failures in hardware, networking, or entire zones. The Application Load Balancer serves as the central traffic manager, enhancing both reliability and scalability.

3. Objectives

- Deploy two EC2 instances, each in a distinct Availability Zone (us-east-1a, us-east-1b).
- Configure a Target Group for logical grouping and health monitoring of EC2 instances.
- Establish an internet-facing Application Load Balancer for efficient HTTP traffic routing.
- Demonstrate round-robin traffic distribution and validate high availability and automatic failover.

4. Architecture Overview

- Compute Layer: Two t2.micro EC2 instances running Amazon Linux 2 and Apache HTTP Server.
- Availability Zones: Instances placed in us-east-1a and us-east-1b.
- Load Balancing Layer: An internet-facing ALB, listening on HTTP port 80, spanning both zones.
- Target Group: Health checks performed on /health.html via HTTP port 80; requires two consecutive successes for healthy status.
- Security Group: Allows inbound SSH (port 22) and HTTP (port 80) from any IP.
- Network: Deployment uses the default VPC and public subnets in both Availability Zones.

5. Results

- Wait 3–5 minutes for ALB to become active.
- Access the ALB DNS name (e.g., http://web-lb-*.us-east-1.elb.amazonaws.com).
- Multiple refreshes showed alternating instance IPs, confirming even traffic distribution across both instances and Availability Zones.
- Health checks functioned correctly, and the setup supports automatic failover.
- Two EC2 instances successfully deployed in separate AZs.
- Target Group monitored both instances as healthy.
- ALB distributed traffic evenly using round-robin.
- High availability achieved: application remains accessible if one instance or AZ fails.

6. Configuration Summary

Component	Details
EC2 Instances	2 (t2.micro, Amazon Linux 2, Apache HTTP Server)
Availability Zones	us-east-1a, us-east-1b
Application Load Balancer	web-lb (internet-facing, HTTP port 80)
Target Group	web-tg (HTTP port 80, /health.html health check, 2 healthy targets required)
Security Group	private-sg (Inbound HTTP 80 & SSH 22 from anywhere)

Network	Default VPC, public subnets in us-east-1a and us-east-1b
Traffic Distribution	Round-robin (stickiness disabled)
Automatic Failover	Supported by ALB health checks and multi-AZ deployment

EC2 INSTANCES

☐	httpserver-2	i-0dd3fb1d0d2180b77	Running	View logs	t2.micro	2/2 checks passed	View alarms	us-east-1b	ec2-3-85-206-78.comp...
☒	httpserver-1	i-09f634fb0045b54dc	Running	View logs	t2.micro	2/2 checks passed	View alarms	us-east-1a	ec2-3-234-5-144.comp...

Target Group

☐	Name	ARN	Port	Protocol	Target type	Load balancer	VPC ID
☐	web-tg	arn:aws:elasticloadbalancin...	80	HTTP	Instance	None associated	vpc-0cf61d4ebb2efce92

Registered targets (2) [Info](#) [Anomaly mitigation: Not applicable](#) [Deregister](#) [Register targets](#)

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

☐	Instance ID	Name	Port	Zone	Health status	Health status details	Administrative ...	Override details	Launch...	Anomal...
☐	i-0dd3fb1d0d2180b77	httpserver-2	80	us-east-1b (us...	Unused	Target group is not co...	-	-	December...	Norm
☐	i-09f634fb0045b54dc	httpserver-1	80	us-east-1a (us...	Unused	Target group is not co...	-	-	December...	Norm

Load Balancer

☐	Name	State	Type	Scheme	IP address type	VPC ID	Availability Zones	Security groups	DNS name
☐	web-lb	Provisioning	application	Internet-facing	IPv4	[REDACTED]	2 Availability Zones	[REDACTED]	web-lb-

Load Balancer with two different EC2 instances

← → ↻ Not Secure http://web-lb-1945963920.us-east-1.elb.amazonaws.com

Import bookmarks... Getting Started

HTTP SERVER 1 with IP

401 - Unauthorized

